



# Ôn tập



## Bài 1

- Cho hệ thống LTI sau:  $y[n] = \frac{1}{2}\{x[n] + x[n - 3]\}$
- Tìm  $y[n]$  ứng với ngõ vào  $x[n] = \begin{cases} 1, & -1 \leq n \leq 1 \\ 0, & n \text{ khác} \end{cases}$
- Tìm đáp ứng tần số  $H(\omega)$  của hệ
- Tính  $|H(\omega)|$

## Bài 2

- Cho hệ thống LTI có impulse response là  $h[n] = \left(\frac{1}{2}\right)^n u[n]$
- Tìm đáp ứng  $y[n]$  ứng với ngõ vào  $x[n] = 1$  khi  $0 \leq n \leq 9$
- Tính  $y[10]$ .





## Bài 1

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ME4083 – Signal processing for mechatronic systems

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$$-1 \leq n-3 \leq 1 \Rightarrow 2 \leq n \leq 4$$

$$\begin{aligned} y[n] &= \frac{1}{2} \{x[n] + x[n-3]\} \\ &= \begin{cases} \frac{1}{2}, & -1 \leq n \leq 1, 2 \leq n \leq 4 \\ 0, & \end{cases} \\ &= \begin{cases} \frac{1}{2}, & -1 \leq n \leq 4 \\ 0, & \end{cases} \end{aligned}$$



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$$x[n] = e^{j\omega n}$$

$$\begin{aligned} y[n] &= \frac{1}{2} (e^{j\omega n} + e^{j\omega(n-3)}) \\ &= \frac{1}{2} e^{j\omega n} (1 + e^{-j3\omega}) \\ &= \frac{1}{2} e^{j\omega n} e^{-j\frac{3}{2}\omega} (e^{j\frac{3}{2}\omega} + e^{-j\frac{3}{2}\omega}) \\ &= e^{j\omega n} e^{-j\frac{3}{2}\omega} \cos\left(\frac{3}{2}\omega\right) \end{aligned}$$





## Bài 1

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$$y[n] = H(\omega)x[n]$$
$$\therefore H(\omega) = e^{-j\frac{3}{2}\omega} \cos\left(\frac{3}{2}\omega\right)$$

$$|H(\omega)| = \left| e^{-j\frac{3}{2}\omega} \right| \left| \cos\left(\frac{3}{2}\omega\right) \right|$$

$$|H(\omega)| = \left| \cos\left(\frac{3}{2}\omega\right) \right|$$

If it is purely complex then you have  $e^{xi} = \cos(x) + i \sin(x)$  the absolute value(  
 $|a + bi| = \sqrt{a^2 + b^2}$ ) is then equal to  $\sqrt{\cos^2(x) + \sin^2(x)} = 1$



## Bài 2

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$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$
$$= \sum_{k=0}^9 h[n-k]$$
$$= \sum_{k=0}^9 \left(\frac{1}{2}\right)^{n-k} u[n-k]$$





## Bài 2

- $n \geq 9$

$$\begin{aligned}y[n] &= \sum_{k=0}^9 \left(\frac{1}{2}\right)^{n-k} \\&= \sum_{k=0}^9 2^{k-n} \\&= \frac{2^{-n}(1 - 2^{10})}{1 - 2} \\&= \left(\frac{1}{2}\right)^{n-10} - \left(\frac{1}{2}\right)^n\end{aligned}$$



## Bài 3

Cho hệ thống rời rạc có hàm truyền như sau:

$$H(z) = \frac{1 - \frac{1}{2}z^{-1}}{1 + \frac{1}{2}z^{-1}}$$

- Tìm  $X(z)$  tương ứng với ngõ vào  $x[n] = \left(\frac{1}{3}\right)^n u[n]$
- Tìm ngõ ra  $Y(z) = H(z)X(z)$





## Bài 3

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$$a^n u[n] \xleftrightarrow{ZT} \frac{1}{1 - az^{-1}}, |z| > |a|$$

$$x(z) = \frac{1}{1 - \frac{1}{3}z^{-1}}, |z| > \frac{1}{3}$$



## Bài 3

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$$Y(z) = H(z)X(z) = \frac{\left(1 - \frac{1}{2}z^{-1}\right)}{\left(1 + \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)}$$

$$Y(z) = \frac{A_1}{\left(1 + \frac{1}{2}z^{-1}\right)} + \frac{A_2}{\left(1 - \frac{1}{3}z^{-1}\right)}$$

$$\left(1 - \frac{1}{2}z^{-1}\right) = A_1\left(1 - \frac{1}{3}z^{-1}\right) + A_2\left(1 + \frac{1}{2}z^{-1}\right)$$

$$\Rightarrow \begin{cases} 1 = A_1 + A_2 \\ -\frac{1}{2} = -\frac{1}{3}A_1 + \frac{1}{2}A_2 \end{cases} \Rightarrow \begin{cases} A_1 = \frac{6}{5} \\ A_2 = -\frac{1}{5} \end{cases}$$





## Bài 4 (Bài 8.5 trang 557)

- The z-transform of the sequence  $x(n) = u(n) - u(n - 7)$  is sampled at five points on the unit circle as follows:

$$X(k) = X(z) \Big|_{z=e^{j2\pi k/5}}$$

- Determine the inverse DFT  $x'(n)$  of  $X(k)$ . Compare it with  $x(n)$  and explain the results.

## Bài 5 (Bài 10.5 trang 736)

- Determine the unit sample response  $\{h(n)\}$  of a linear-phase FIR filter of length  $M = 4$  for which the frequency response at

$$\omega = 0 \text{ and } \omega = \frac{\pi}{2} \text{ is specified as } H_r(0) = 1, H_r\left(\frac{\pi}{2}\right) = \frac{1}{2}$$



## Bài 4

$$X(z) = 1 + z^{-1} + \dots + z^{-6}$$

$$\begin{aligned} X(k) &= X(z) \Big|_{z=e^{j\frac{2\pi}{5}}} \\ &= 1 + e^{-j\frac{2\pi}{5}} + e^{-j\frac{4\pi}{5}} + \dots + e^{-j\frac{12\pi}{5}} \\ &= 2 + 2e^{-j\frac{2\pi}{5}} + e^{-j\frac{4\pi}{5}} + \dots + e^{-j\frac{8\pi}{5}} \\ x'(n) &= \{2, 2, 1, 1, 1\} \end{aligned}$$

- Temporal aliasing occurs in first two points of  $x'(n)$  because  $X(z)$  is not sampled at sufficiently small spacing on the unit circle.





## Bài 5

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$$M = 4, \quad H_r(0) = 1, \quad H_r\left(\frac{\pi}{2}\right) = \frac{1}{2}$$

$$\begin{aligned} H_r(w) &= 2 \sum_{n=0}^{\frac{M}{2}-1} h(n) \cos\left[w\left(\frac{M-1}{2} - n\right)\right] \\ &= 2 \sum_{n=0}^1 h(n) \cos\left[w\left(\frac{3}{2} - n\right)\right] \end{aligned}$$



## Bài 5

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$$\text{At } w = 0, H_r(0) = 1 = 2 \sum_{n=0}^1 h(n) \cos[0]$$

$$2[h(0) + h(1)] = 1 \quad (1)$$

$$\text{At } w = \frac{\pi}{2}, H_r\left(\frac{\pi}{2}\right) = \frac{1}{2} = 2 \sum_{n=0}^1 h(n) \cos\left[\frac{\pi}{2}\left(\frac{3}{2} - n\right)\right]$$

$$-h(0) + h(1) = 0.354 \quad (2)$$

Solving (1) and (2), we get

$$h(0) = 0.073 \text{ and}$$

$$h(1) = 0.427$$

$$h(2) = h(1)$$

$$h(3) = h(0)$$

$$\text{Hence, } h(n) = \{0.073, 0.427, 0.427, 0.073\}$$